Topology-Aware Mapping for Photonic Interconnects in TPU-Class LLM Accelerators

Technical Summary Sheet

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I. Key Insights and Innovations

1) **The intra-chip mapping is invariant under photonics; the distributed schedule is what changes.** For the 64-chip TPU cube, AccelForge converges to the same partitioning, dataflow, and tile signature for all seven modeled interconnect technologies (0/24 inner mappings change). The benefit lives one layer up: 5/16 workload–scenario pairs flip their optimal communication pattern, and switching to remote pooled-HBM streaming on EVK100 removes 73.86% of compute energy on KV attention. Reporting these as separate layers matches the 6.5930 mapping formalism rather than collapsing them into a single "mapping changed" number.
2) **Hardware multicast and distance-insensitive transport, not raw bandwidth, dominate shared-state workloads.** A 3.5 Tb/s, 6 pJ/bit advanced chiplet baseline—chosen specifically to defeat the bandwidth hypothesis—is still beaten by Lightmatter M1000 by 99.85% on shared-DRAM multicast: N unicasts collapse to one logical broadcast and contention disappears on a switched fabric. Our distance-parametrized link model splits each technology into a conversion term and a per-mm transmission term; electrical pJ/bit grows with distance while photonic pJ/bit stays flat, and a regression test locks the crossover in.
3) **Photonics is not a uniform multiplier.** On the 9216-chip Ironwood-class superpod, the public Gemma 4 26B MoE proxy reaches 42.43× inference latency speedup and 94.05% energy reduction with the full M1000 photonic stack; the larger long-context Gemini Deep Think proxy reaches 14.50×. Training reaches only 3.37× and 1.43× respectively. Strong scaling exposes the compute/HBM Amdahl ceiling; weak scaling shows where photonic multicast still pays. The strong case for photonics is inference and long-context serving, not training payoff.

II. Contributions

Simone Surdi. AccelForge / network model: topology classes (ring, mesh, 3-D torus, optical crossbar / switched fabric); the `LogicalClusterMemory` extension; the iterative network-aware feedback loop (`solve_iteratively`); mapping-signature extraction; cluster, TPU-cube, and chiplet-baseline studies; technology-catalog constants and sensitivity sweeps.

Pranav Sitaraman. Compiler / superpod analysis: distance-parametrized link model; OCS/OPS sensitivity; the four named compiler-visible distributed schedules (SPMD collective, remote pooled-HBM streaming, replicated cache, photonic multicast cache fill); the 9216-chip Ironwood-class stage model (prefill / decode / refill / MoE / training / gradient / optimizer); strong and weak scaling on Gemma 4 anchors plus Gemini-class stress proxies; ablation ladder; integration tests; and the report, poster, technical summary, and presentation deliverables.

III. Artifact

Repository. `final-project/workspace/final_project/` in the course repository (extends the AccelForge cluster example). **Reproduction.** `cd final-project && make study && make notebook && make test && make deliverables` runs inside the `accelforge:latest-amd64` Docker image. **Deliverables.** `paper.pdf` (4-page IEEE conference report, grad version), this technical summary (`technical_summary.pdf`), `poster.pdf` (a0 portrait), and `presentation.pdf` (beamer slides). **Engine.** `scripts/lightmatter_interconnects.py` holds the technology catalog, topology classes, iterative solver, distance physics, strong/weak and OCS/OPS studies, and the superpod stage model. **Notebook.** The main notebook is regenerated from the checked-in generator and executed end-to-end as part of validation. **Test suite.** `tests/test_lightmatter_project.py` is the canonical correctness check and includes regression tests for the electrical/photonic distance crossover, the mapping-invariance claim, the multicast scaling exponent, the strong/weak modes, and the OCS/OPS sensitivity. **Figures.** The six figures and compact assumption/result tables in `results/presentation/` are the canonical final submission package; `results/` also holds the raw CSV artifacts that back every quantitative claim in this summary.